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Zahra Shakeri, Marwa Sharifi, Sepehr Harfi Moridani, Chan Pang Yang, Konrad Samsel, Christoffer Dharma & Mohammad Noaeen

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Detecting Misconception Surges in Opioid Policy: A Zero-Shot LLM Analysis of British Columbia's Decriminalization Pilot

Zahra Shakeri^{1,2,3*}, Marwa Sharifi¹, Sepehr Harfi Moridani¹, Chan Pang Yang¹, Konrad Samsel¹, Christoffer Dharma^{1*}, and Mohammad Noaeen¹

¹Institute of Health Policy, Management and Evaluation, Dalla Lana School of Public Health, University of Toronto, Toronto, Canada

²Faculty of Information, University of Toronto, Toronto, Canada

³Schwartz Reisman Institute, University of Toronto, Toronto, Canada

*Corresponding author, chris.dharma@mail.utoronto.ca, zahra.shakeri@utoronto.ca

Abstract

Background: In January 2023, British Columbia (BC) launched Canada's first drug decriminalization pilot in response to persistently high opioid-related deaths. This initiative removed criminal penalties for possession of up to 2.5 grams of certain illicit drugs to reduce stigma and increase service use. Social media platforms such as Reddit provide a real time window into public perceptions and misconceptions about such policies.

Methods: We analyzed 12,537 Reddit comment records posted between January 2023 and October 2024 from subreddits discussing decriminalization in British Columbia. A zero-shot large language model applied a multi-label taxonomy with four misconception classes and one no-claim class. Expert review of 547 comments with model disagreement supported selection of the primary classifier for analysis. We tracked monthly misconception prevalence and used subreddit-week reply premiums and lattice clustering to detect engagement bursts. Negative binomial models quantified reply differences across periods, regions, and comment-level harm reduction references.

Results: GPT-4o assigned at least one substantive misconception label to 2,457 comment records (19.60%). Label 5 was assigned to 10,080 records (80.40%), consistent with a conservative taxonomy that separates checkable policy claims from opinion, sarcasm, anecdote, and moral judgment. After the May 2024 amendment that restricted possession in public spaces, legal status and enforcement misconceptions increased. Legal or enforcement misconception records had a higher adjusted reply-rate estimate than other records, although the main effect had limited precision (IRR 1.13, 95% CI 1.00-1.27, $p = 0.059$). The post-May interaction was positive (IRR 1.14, 95% CI 1.02-1.27, $p = 0.019$). Reply premiums clustered into three premium bursts from late March to mid-May 2024. Harm reduction references had higher adjusted replies (IRR 1.14, 95% CI 1.07-1.22, $p < 0.001$), while co-occurrence of harm reduction references with legal or enforcement misconceptions was associated with a smaller added reply premium (IRR 0.80, 95% CI 0.66-0.97, $p = 0.022$).

Conclusions: Public discussion of BC's decriminalization policy revealed persistent confusion, and legal and enforcement confusion intensified around the May 2024 revision. Burst detection on social platforms can identify when and where confusion concentrates, which supports timely public clarification and service signposting. During future amendments, concise messages about permissible locations and remaining police authority may reduce avoidable enforcement contact and misinformation.

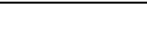
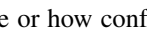
Keywords: Decriminalization, Drug policy, Social listening, policy misconceptions, Harm reduction communication, Large language models

1 Introduction

The opioid crisis is a major public health challenge in Canada and has caused over 50,000 deaths since 2016.¹ In 2024 alone, 7,146 opioid related deaths were reported, an average of 20 deaths per day.¹ Opioids include prescription analgesics such as morphine and oxycodone as well as illicit substances like fentanyl and heroin.^{2,3} The route of administration, frequency of use, and individual tolerance levels all affect the risk of harm.^{4,5} Co-occurring mental health conditions and limited access to harm reduction services can increase the likelihood of dependence or overdose.⁶ Opioid use imposes an estimated annual economic burden of \$7.1 billion in Canada, in-

cluding substantial healthcare and emergency response costs.^{7,8} In 2020, opioid-related healthcare costs alone were \$519 million, within the \$13.4 billion attributed to substance use in the Canadian healthcare system.⁷ Despite significant investments in healthcare and prevention, opioid toxicity deaths continue to be reported at high levels, with the highest counts in British Columbia (BC), Alberta (AB), and Ontario (ON).¹ Many factors contribute to the ongoing opioid crisis and poor patient outcomes, with stigma being a significant one. Stigma can discourage individuals from seeking help due to fear of judgment, and often results in lower quality care when accessing services.⁹ A qualitative study conducted in BC found that people who use drugs usually face discrimination in healthcare settings,

Table 1: **Characteristics of Reddit comment records related to decriminalization in British Columbia by subreddit.** The rightmost column shows a miniature filled line of monthly comment counts with a shared vertical scale across rows. All densities cover 2023-01-31 to 2024-10-31. The hosted export contained 22,132 rows. After one incomplete row was excluded, exact repeated export rows were collapsed to one comment record, which gave a final analytic corpus of 12,537 comment records.

Subreddit	Comments	Total replies	Mean replies (SD)	Total upvotes	Median upvotes [IQR]	Date range	Density (monthly comments)
r/canada	5,401	4,292	0.8 (1.1)	48,680	2 [1–7]	2023-02-22 to 2024-09-08	
r/vancouver	1,833	1,421	0.8 (1.1)	27,957	4 [1–13]	2023-05-14 to 2024-10-20	
r/britishcolumbia	1,265	922	0.7 (1.1)	13,637	2 [1–6]	2023-01-31 to 2024-08-04	
r/Canada_sub	879	620	0.7 (0.9)	2,033	1 [1–2]	2023-07-19 to 2024-07-27	
r/VictoriaBC	757	597	0.8 (1.1)	5,284	2 [1–6]	2023-08-13 to 2024-08-30	
r/onguardforthee	597	395	0.7 (1.0)	6,194	3 [1–9]	2023-02-22 to 2024-10-08	
r/CanadaPolitics	593	532	0.9 (1.2)	4,424	3 [1–8]	2024-04-22 to 2024-07-30	
r/ontario	480	387	0.8 (1.2)	3,713	2 [1–5]	2024-03-30 to 2024-06-07	
r/toronto	314	250	0.8 (1.0)	2,425	2 [1–8]	2024-05-01 to 2024-05-20	
r/BCpolitics	157	118	0.8 (0.7)	392	1 [0–3]	2023-05-18 to 2024-09-18	
r/saskatoon	153	121	0.8 (1.1)	801	1 [1–6]	2024-04-29 to 2024-05-22	
r/saskatchewan	72	56	0.8 (0.9)	462	3 [1–6]	2024-04-29 to 2024-05-01	
r/UBC	33	20	0.6 (0.7)	314	4 [1–11]	2023-02-01 to 2023-02-03	
r/NiceVancouver	3	0	0.0 (0.0)	4	1 [1–2]	2023-08-14 to 2023-08-15	
All subreddits	12,537	9,731	0.8 (1.1)	116,320	2 [1–7]	2023-01-31 to 2024-10-20	

are denied pain medications, and are labeled as 'drug-seeking'. These experiences can lead to self-medication with illicit substances and increase overdose risk.¹⁰ In response, several regions worldwide have considered or implemented decriminalization to reduce stigma and encourage treatment.^{11,12} Oregon's Measure 110 decriminalized possession of small amounts of specified drugs in 2021.¹³ The state reinstated criminal penalties in September 2024 after concerns arose about policy design and implementation.¹⁴ In Canada, British Columbia became the first province to implement a drug decriminalization pilot. On 31 January 2023, Health Canada granted BC a three year exemption that removed criminal penalties for adults possessing up to 2.5 grams of selected illicit substances for personal use in settings covered by the exemption.¹⁵ The policy aimed to reduce stigma and connect people to health and social services instead of criminal penalties.¹⁶ The exemption did not legalize drug production, sale, trafficking, or impaired driving, and it did not apply in all settings. In May 2024, the exemption was amended so that possession in public spaces was no longer covered. It then applied only in private residences, places where a person was legally sheltering, supervised consumption sites, and overdose prevention sites.

Early evidence has also begun to document how BC's decriminalization pilot and its May 2024 amendment have been experienced in frontline service settings. Using key-informant interviews with harm reduction and opioid agonist treatment providers across the province, Russell et al. (2026) reported few immediate operational changes after decriminalization but identified persistent resource constraints and substantial uncertainty related to limited policy communication and site-specific training, mainly around the May 2024 revision that recriminalized possession in public spaces.¹⁷ These implementation experiences suggest that decriminalization's public health impact depends not only on the legal change itself, but also on how clearly the policy is communicated and understood across stakeholders. However, the study did not address how the wider

public interpreted policy details in real time or how confusion shifted during policy revisions, especially in large and high intensity online discussions.

Public understanding can define how decriminalization is received and if people at risk feel safer seeking help. Public discussion can also spread inaccurate claims about the policy's boundaries, including claims that conflate decriminalization with legalization or misstate police authority under the exemption.^{18,19,17} These errors can reduce trust, create uncertainty among people who use drugs, and make communication harder during policy amendments. Social media platforms such as Reddit provide time stamped discussions during policy change.²⁰ Unlike traditional surveys, which can be limited by recall bias or social desirability effects,^{21,22} Reddit discussions often include candid reflections on drug use, harm reduction, and policy outcomes.²⁰ This makes Reddit useful for studying where and when policy confusion appears during amendment periods, when legal boundaries can shift quickly.

In this study, we address three key questions. First, what are the prevalent misconceptions about BC's decriminalization policy expressed in online discussions, and how did the prominence of these themes change over time? Second, do comments containing misconceptions about the policy's legal status or enforcement attract unusually high levels of engagement, and if so, when do these intense discussions ('bursts') occur? Third, do these comments containing misconceptions have higher levels of engagement when there is more visible harm reduction content in a given community? By answering these questions, we aim to explore *what* the public tends to misconceive about the policy, *when* these misunderstandings generate surges of engagement, and *how* public engagements in these misconceptions are potentially modified with increased harm reduction messaging. Clarifying these aspects can inform targeted communication strategies. For instance, knowing which specific misunderstandings to address and when to intervene (e.g., during intense online debates or after policy updates) can help pub-

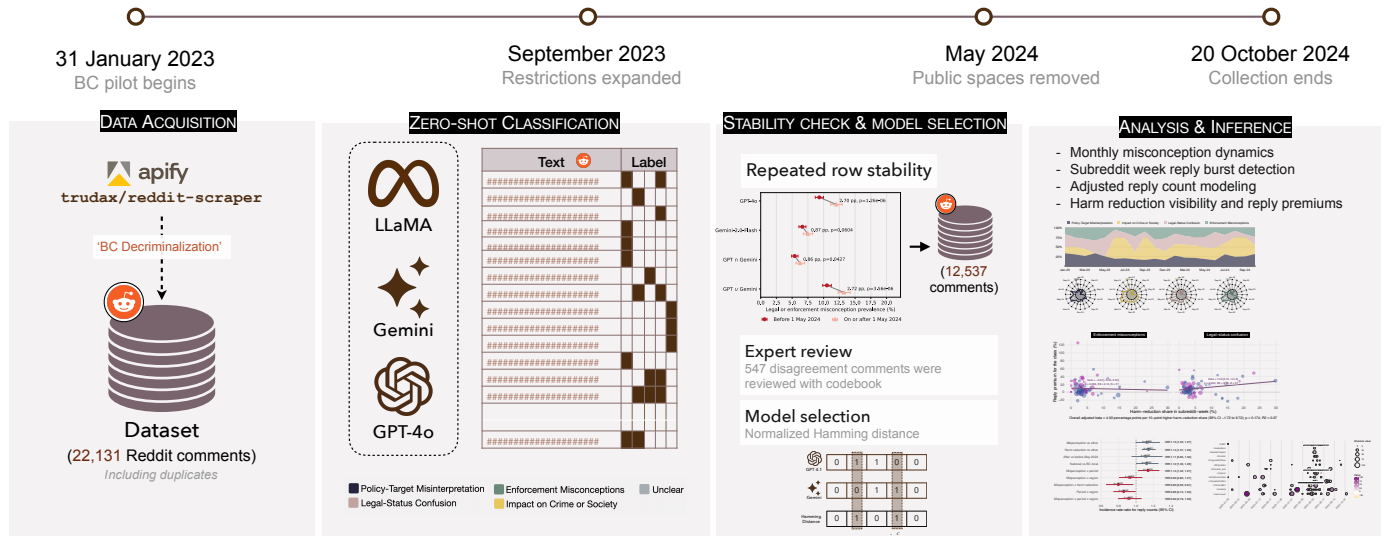


Figure 1: **Overall study workflow.** Reddit export rows were cleaned into a comment level corpus, classified with zero-shot multi-label models, checked for stability, and reviewed against expert labels for model selection. The final labels supported monthly misconception trends, reply burst detection, reply count modeling, and harm reduction visibility analysis.

lic health officials focus their communication efforts where they are most needed.

2 Materials and Methods

Figure 1 summarizes the full study workflow. Reddit export rows were cleaned into comment records, classified with zero-shot multi-label models, checked for stability, reviewed for model selection, and used for monthly trend, reply burst, reply count, and harm reduction analyses.

2.1 Data Collection

We constructed a comment-level Reddit corpus to study public discourse following the implementation of British Columbia's drug decriminalization policy, effective 31 January 2023. Data acquisition used a hosted third party interface on Apify (i.e., *trudax/reddit-scraper*). The Apify actor was configured with the case insensitive search phrase 'BC Decriminalization' and exported matched Reddit threads and comments to CSV. Post titles and thread metadata were used to identify relevant discussions and organize comments, but post bodies, self post text, link post titles, linked article content, and article comment sections were not treated as analytic observations. We focused on comments as both study outcomes (i.e., misconception labels and direct replies) were measured at the comment level. Self posts and link posts initiate discussions, while comments capture within thread exchanges. Combining those units in one reply model would mix thread initiation with response behaviour and would require a different analytic design.

The hosted export contained 22,132 rows. One row lacked subreddit, timestamp, direct reply count, and upvote values and was excluded. The remaining export rows represented 12,537 distinct comment records after exact repeated export rows were

collapsed by identical comment text, subreddit, timestamp, direct reply count, and upvote count after numeric fields were standardized. We did not merge near-duplicate comments or comments that shared text but differed by time, subreddit, direct replies, or upvotes. Exact repeated rows were used only to form one comment-level record and to check label stability across repeated model outputs, as described below. Each final comment record includes raw comment text, number of upvotes, publication timestamp, subreddit name, and direct reply count. Usernames, user identifiers, profile links, and account histories were not kept in the analytic files and were not used in any analysis.

Table 1 summarizes the final analytic corpus of 12,537 comment records from January 2023 to October 2024. For each subreddit, the table reports the number of comment records, total replies, mean replies per record, total upvotes, median upvotes with interquartile range, calendar coverage, and monthly comment density on a shared vertical scale. Activity was concentrated in *r/canada* (5,401, 43.08%), *r/vancouver* (1,833, 14.62%), and *r/britishcolumbia* (1,265, 10.09%). The subreddits shown in the table are the communities present in the archived export returned by the configured query at the time of collection. We did not manually add, remove, or screen communities after export. The corpus should be read as a Reddit sample based on the phrase 'BC Decriminalization', not as a Reddit wide census of all discussions about decriminalization in British Columbia. Across all subreddits, records had a mean of 0.8 direct replies (SD 1.1) and a median of 2 upvotes (IQR 1 to 7). The dataset used in this study and the code to reproduce the pipeline are available for replication and further analysis.²³

2.2 Policy Misconception Classification

We defined *policy misconception* as a checkable factual error about the intent, legal scope, enforcement practice, or al-

leged downstream impacts of BC's decriminalization exemption. We built a five-class taxonomy to balance policy relevance and labeling reliability. Four classes captured misconception types that recur in public debate and appear in prior work on decriminalization knowledge gaps, definitional confusion, and policy framing.^{24,25,26,18} A fifth class captures comments that contain no verifiable factual claim and cannot be classified as correct or incorrect. We kept the taxonomy small to support consistent annotation at scale and clear interpretation for a public health audience. Table 2 lists these labels, along with the operational definition we applied during labeling and an illustrative example for each class. We designed the classes to map onto concrete policy elements that are often misunderstood: the policy purpose (Label 1), the difference between decriminalization and legalization (Label 2), enforcement powers and limits (Label 3), and causal claims about crime or disorder (Label 4). We used Label 5 only when the comment contained no checkable factual statement about the policy. To apply this taxonomy, we used the policy rules in place at the time each comment was posted as the reference point. We then separated claims about the legal framework, enforcement authority, and policy outcomes. Legal framework claims were evaluated against what the exemption allowed or excluded. Enforcement claims were evaluated against police authority to seize drugs, intervene, or act on trafficking. Outcome claims were assigned Label 4 only when the comment presented the exemption as the direct, sole, or settled cause of crime, overdose deaths, or public disorder. Comments that expressed concern, criticism, support, sarcasm, moral judgment, personal experience, or prediction without a checkable policy claim were assigned Label 5.

Since many comments can contain more than one checkable policy claim, we used a multi-label format. The prompt asked the model to use fewer labels when evidence was weak and to avoid assigning extra labels for tone or political stance alone. Parser logs were checked for outputs that exceeded the requested cap. In the GPT-4o analytic file, this occurred for one record. Downstream analyses used binary indicators for each class, so label order was not used. We used zero-shot multi-label classification, where the model assigns labels from Table 2 without task specific training examples.^{27,28} This choice matched our goal of measuring how general purpose models interpret policy claims in public discussion, and it avoided a custom training set tied to one period's phrasing.

During parsing, Label 5 was treated as a mutually exclusive no claim category. If a model returned Label 5 together with any of Labels 1 to 4, Label 5 alone was removed and the substantive labels were kept. This step did not delete the comment and did not add a substantive label. It also did not assign lower trust to Label 5. It only enforced the codebook rule that Label 5 applies when no checkable policy claim is present. For parsed model output L_j , the cleaned label set was

$$L_j^* = \begin{cases} L_j \setminus \{5\}, & \text{if } L_j \cap \{1, 2, 3, 4\} \neq \emptyset, \\ \{5\}, & \text{if } L_j \cap \{1, 2, 3, 4\} = \emptyset. \end{cases}$$

For a repeated row group g with n_g export rows and model m , the final binary label for each substantive class $c \in \{1, 2, 3, 4\}$

was defined as $\tilde{y}_{g,c}^{(m)} = \mathbf{1} \left\{ \sum_{i \in g} y_{i,c}^{(m)} > \frac{n_g}{2} \right\}$. The final label set was $\tilde{L}_g^{(m)} = \{c \in \{1, 2, 3, 4\} : \tilde{y}_{g,c}^{(m)} = 1\}$. If $\tilde{L}_g^{(m)}$ was empty, the final label set was assigned Label 5: $\tilde{L}_g^{(m)} = \{5\}$.

This rule prevented repeated export rows from contributing more than once to prevalence estimates, monthly patterns, burst detection, or reply models. It also avoided treating a single unstable repeated output as enough evidence for a substantive misconception label. The repeated row groups also provided a direct check of label stability. Across 9,578 repeated row groups, GPT-4o returned the same parsed label set in 9,057 groups (94.56%) and a different parsed label set in 521 groups (5.44%). Gemini-2.0-Flash returned the same parsed label set in 9,244 groups (96.51%) and a different parsed label set in 334 groups (3.49%). LLaMA-3.2B returned the same parsed label set in 3,057 groups (31.92%) and a different parsed label set in 6,521 groups (68.08%). These repeated-output checks matched the broader model comparison, where GPT-4o and Gemini-2.0-Flash were stable under the taxonomy, while LLaMA-3.2B changed labels more often for the same record.

To label policy misconceptions at scale and maintain comparability across weeks and subreddits, we used the following fixed and taxonomy-based prompt with GPT-4o and Gemini-2.0-Flash.^{29,30} The prompt returned only valid label IDs, separated checkable policy claims from rhetorical opinion, and kept the label boundaries stable under slang changes and event-driven spikes. Model outputs were parsed as unordered label sets. For comment j , the parsed output was represented as $L_j \subseteq \{1, 2, 3, 4, 5\}$. The model did not return confidence scores, and labels were not ranked by confidence, output order, or position of the claim in the comment text. Downstream analyses used binary class indicators $\mathbf{1}(c \in L_j)$, so label order had no role in prevalence estimates, burst detection, or reply models.

Policy Misconceptions Prompt

You are a public health policy analyst. The task is to identify up to three factual error types in a Reddit comment about British Columbia's drug decriminalization pilot. Select labels from this list:

- 1. Policy-Target Misinterpretation: The comment misrepresents the purpose of the policy. For example, it claims the policy promotes drug use rather than access to health and social services.
- 2. Legal-Status Confusion: The comment conflates decriminalization with legalization. For example, it claims drugs are fully legal or permitted everywhere.
- 3. Enforcement Misconceptions: The comment misstates police or authority powers. For example, it claims police cannot intervene, confiscate drugs, or act on trafficking.
- 4. Impact on Crime or Society: The comment attributes crime, overdose deaths, or disorder directly to the policy without credible support.
- 5. Unclear or no verifiable factual claim: Use this label only when no other label applies.

Apply these rules:

- The classification should focus on factual claims rather than tone or opinion.
- The output should include one to three labels. Use fewer labels when the evidence is weak.
- Label 5 applies to satire, unverifiable anecdotes, or comments without a checkable factual claim.

Output the label numbers only, separated by commas.

Comment: [Comment Body]

Table 2: **Policy misconception taxonomy used for multi-label classification.** Model outputs were parsed as unordered label sets, and each comment could receive more than one substantive label when more than one checkable claim was present. Label 5 was treated as a mutually exclusive no claim category after parsing. Examples are illustrative and paraphrased.

Label	Operational definition (what we label)	Illustrative example
1. Policy-Target Misinterpretation	The comment misstates the purpose of the exemption. It presents the policy as promoting or encouraging drug use, rather than reducing harms by shifting personal possession away from criminal penalties and toward health and social supports.	“They want everyone to start using drugs.”
2. Legal-Status Confusion	The comment conflates decriminalization with legalization or regulation. It claims drugs are “legal”, can be bought legally, or can be used anywhere without restriction.	“Drugs are legal now, so you can do them anywhere.”
3. Enforcement Misconceptions	The comment misstates what police or other authorities can do. It claims enforcement has stopped entirely, that police cannot seize drugs, or that trafficking is permitted or ignored.	“Police cannot take drugs or stop dealers anymore.”
4. Impact on Crime or Society	The comment asserts a direct, sole, or definitive causal link from the exemption to changes in crime, overdoses, or public disorder without credible support. We applied this label to causal claims framed as settled fact rather than as uncertainty or personal experience.	“Crime exploded because of decriminalization, nothing else.”
5. Unclear or no verifiable factual claim	The comment contains no checkable factual claim about the policy. It is pure opinion, sarcasm, insult, or anecdote without a falsifiable statement about intent, scope, enforcement, or impacts. We used this label only when none of Labels 1-4 applied.	“This policy is a disaster”.

To compare label sources under a multi-label setup, we represented each comment as a binary vector over the five labels, with $y_{j,c} = 1$ when class c was assigned to comment j . We then used mean normalized Hamming distance:^{31,32}

$$H(A, B) = \frac{1}{NC} \sum_{j=1}^N \sum_{c=1}^C \mathbf{1}\{y_{j,c}^A \neq y_{j,c}^B\},$$

	Label 1	Label 2	Label 3	Label 4	Label 5
GPT-4o	0	1	1	0	0
Gemini-2.0-Flash	0	0	1	1	0
Mismatch	0	1	0	1	0

Mismatch count = 2
Normalized Hamming = 2/5 = 0.40

For one comment: $H(A, B) = \frac{1}{C} \sum_{c=1}^C \mathbf{1}\{y_c^A \neq y_c^B\}$

where $\mathbf{1}\{\cdot\}$ is the indicator function, (N) is the number of comments, and $(C=5)$ is the number of labels. This metric gives the average share of label positions on which two sources disagree. We selected this metric to account for comments receiving more than one label, since a partial mismatch should not be treated the same as a completely different label set. We also tested LLaMA-3.2B as a lightweight open source baseline during early pilot labeling.^{33,34} After comment level label construction, GPT-4o and Gemini-2.0-Flash had close agreement, with pairwise normalized Hamming distance 0.0590. LLaMA-3.2B differed more from GPT-4o and Gemini-2.0-Flash, with pairwise distances 0.3227 and 0.3266, respectively. LLaMA-3.2B also changed labels more often across repeated outputs for the same comment record. These results led us to use LLaMA-3.2B only as a baseline model.

For final model selection, two domain experts reviewed the 547 comments on which GPT-4o and Gemini-2.0-Flash disagreed during the initial labeling pass. Both experts applied the five class definitions in Table 2 and agreed on a reference label set for these cases. We refer to this as the expert reference label set to distinguish it from a corpus wide ground truth, which the review was not intended to provide. We compared each model output with the expert reference labels using

the normalized Hamming distance defined above. GPT-4o had a lower average disagreement than Gemini-2.0-Flash and was selected as the primary classifier. The review was limited to model selection on disputed cases. It did not measure blinded interrater reliability, estimate corpus wide accuracy, or involve prompt tuning. Neither model was tuned on the review set.

2.3 Reply Burst Detection

Reply premium and hot cells. Our second research question focused on which policy misconceptions triggered unusually high reply activity and when these surges occurred. We focused on the ‘Legal-Status Confusion’ and ‘Enforcement Misconceptions’ categories, which directly concern the legal scope of the exemption and police powers. To compare replies to misconception comments against replies to other comments in the same subreddit and week, we measured burst-like engagement³⁵ at the subreddit-week level using a *reply premium*. If r_i denotes the direct reply count for comment i , we define the variance-stabilized outcome as

$$y_i = \log(1 + r_i).$$

For each subreddit s and calendar week w , we define $M_i = 1$ to indicate that comment i contains either a legal-status misconception or an enforcement misconception, and we define $M_i = 0$ otherwise. We compute the within-cell mean difference

$$\Delta_{s,w} = \bar{y}_{s,w}^{(1)} - \bar{y}_{s,w}^{(0)}, \quad \bar{y}_{s,w}^{(m)} = \frac{1}{N_{s,w}^{(m)}} \sum_{i \in (s,w): M_i=m} y_i,$$

where $N_{s,w}^{(m)}$ is the number of comments in subreddit s and week w for which $M_i = m$. We then map this difference to a percentage scale:

$$\text{Premium}_{s,w} = 100 \times \{\exp(\Delta_{s,w}) - 1\}.$$

A positive value indicates higher reply engagement for misconception comments than for other comments within that same

subreddit-week. The subreddit-week lattice preserved the full calendar sequence of weeks in the collection window. Weeks with undefined premiums (i.e., $N_{s,w}^{(1)} = 0$ or $N_{s,w}^{(0)} = 0$) were left empty instead of removed, so burst detection used adjacency in calendar time. Since reply premiums vary across subreddits, we standardized each subreddit's weekly series. For each subreddit s , we computed

$$z_{s,w} = \frac{\text{Premium}_{s,w} - \overline{\text{Premium}_s}}{\text{SD}(\text{Premium}_{s,\cdot})},$$

where the mean and standard deviation were computed over weeks with defined premiums. We then defined sign-specific 'hot' thresholds using quantiles of the standardized distribution, where the positive cutoff z^+ was the q th quantile of $z_{s,w} : z_{s,w} > 0$ and the negative cutoff z^- was the q th quantile of $-z_{s,w} : z_{s,w} < 0$, with $q = 0.85$. A subreddit-week was flagged as a candidate *premium* hot cell if

$$z_{s,w} \geq z^+ \quad \text{or} \quad \text{Premium}_{s,w} \geq a,$$

and as a candidate *penalty* hot cell if

$$z_{s,w} \leq -z^- \quad \text{or} \quad \text{Premium}_{s,w} \leq -a,$$

with an absolute cutoff of $a = 10$ percentage points. This combined rule captures weeks that are extreme in relation to the typical variability of a subreddit and weeks that are practically large on the original percentage scale.

Spatiotemporal clustering into bursts. Next, we treated each flagged subreddit-week as a grid cell and formed clusters using 8-neighbor adjacency, where cells touch by an edge or a corner. Here, 'spatial' refers to adjacency on the subreddit-week lattice (i.e., subreddit index $\times$ time), not geographic proximity. We computed connected components separately for premium and penalty cells. Each connected component was summarized by (i) the number of hot cells, (ii) its week span, (iii) its subreddit span, and (iv) a coherence measure defined as the fill ratio of hot cells within the component's axis-aligned bounding box:

$$\text{Fill} = \frac{N_{\text{cells}}}{(\text{weeks span}) \times (\text{subreddits span})}$$

We kept bursts that spanned at least 2 weeks, involved at least 2 subreddits, contained at least 4 hot cells, and had $\text{Fill} \geq 0.35$. These requirements removed isolated spikes and focused the analysis on engagement surges that lasted for weeks and appeared in more than one subreddit. We then plotted the weekly reply premium surface and overlaid the qualifying connected components. This procedure produced the subreddit by week burst map shown in Figure 5.

2.4 Harm reduction discourse and engagement with legal/enforcement misconceptions

To understand how harm reduction discourse defines engagement with legal status and enforcement misconceptions about BC's decriminalization policy, we analyzed how comments that misstate the scope of the exemption or police powers

perform in terms of replies and how this engagement varies in weeks when harm reduction content is more visible in a given community. To this end, we started with the comment-level Reddit corpus described above. Each row represented one comment and included raw text, subreddit, timestamp, direct reply count, upvotes, and the multi-label misconception string. The exports used different column names, so we standardized the headers and mapped each field to the first matching candidate. Header names were converted to lowercase and spaces were removed. Only the selected columns were kept. Timestamps were parsed from several human readable formats and from epoch values (e.g., seconds or milliseconds). All times were converted to UTC to create the week, hour-of-day, and week-day variables. Rows were removed if the subreddit, timestamp, reply count, or text was missing.

We defined BC focused communities as subreddits such as *r/vancouver*, *r/britishcolumbia*, *r/VictoriaBC*, *r/BCpolitics*, and *r/UBC*. These subreddits were coded as region *BC_local*, while all other subreddits were coded as region *National*. Indicator *post* equaled 1 on or after 1 May 2024 and 0 before, and comment text and label string were converted to lower case before pattern matching.

Harm reduction and misperception indicators. We identified explicit harm reduction content using a rule based approach.^{36,37} After converting text to lower case, we searched for terms related to overdose response, drug checking, supervised consumption and injection sites, overdose prevention sites, syringe service programs, opioid agonist therapies, prescribed safer supply, and key provincial institutions. Comments containing at least one matched term were coded as containing harm reduction content. To align the indicator with the comment-level reply outcome, we derived it only from the text of the comment that directly received replies. Post-level material, including self-post text and link-post titles, and externally linked content, including articles and their comment sections, were outside the analytic unit. The existing label string encoded four misconception classes. For this section, we focused on comments that misrepresented the legal status of possession and use or the powers of police and other authorities. We created a combined indicator for legal and enforcement misconceptions that was set to one when either of these classes was present and zero otherwise.

Comment-level model of reply counts. Let Y_i denote the number of direct replies to comment i . Reply counts were modelled with a negative binomial regression and a log link,

$$Y_i | X_i \sim \text{NB}(\mu_i, \theta), \quad \log(\mu_i) = X_i^T \beta.$$

The core predictors were M_i , Post_i , and Region_i . The policy-period indicator Post_i identified comments on or after May 1, 2024, while Region_i distinguished BC local from national subreddits. The specification included all two-way interactions and the three-way interaction among these predictors. The harm reduction indicator HR_i was included as a main effect and through its interaction with M_i , written as $M_i \times \text{HR}_i$. The adjustment set comprised $\log(1 + \text{len}_i)$, upvotes, hour of day, weekday, and subreddit fixed effects. If the negative bino-

Table 3: **Detailed model outputs after comment-record construction and conservative comment-level label assignment.** *Avg. Labels* denotes the average number of labels assigned per comment record. *Mean Dist.* denotes the average of a model’s pairwise normalized Hamming distances to the other two models.

Model	L1	L2	L3	L4	L1-L4 (Any)	L5	Avg. Labels	Mean Dist.
LLaMA-3.2B	4,619 (36.84%)	1,990 (15.87%)	2,943 (23.47%)	4,856 (38.73%)	7,864 (62.73%)	4,673 (37.27%)	1.52	0.3247
GPT-4o	649 (5.18%)	776 (6.19%)	694 (5.54%)	961 (7.67%)	2,457 (19.60%)	10,080 (80.40%)	1.05	0.1909
Gemini-2.0-Flash	280 (2.23%)	446 (3.56%)	588 (4.69%)	1,207 (9.63%)	1,989 (15.87%)	10,548 (84.13%)	1.04	0.1928

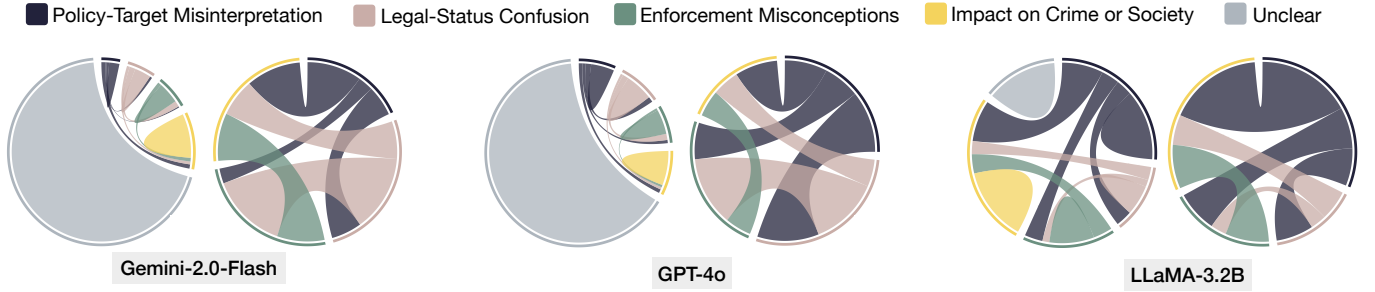


Figure 2: **Thematic Label Distribution and Co-occurrence Analysis.** Chord diagrams comparing the classification behaviour of Gemini-2.0-Flash, GPT-4o, and LLaMA-3.2B across the five thematic categories: Policy-Target Misinterpretation, Legal-Status Confusion, Enforcement Misconceptions, Impact on Crime or Society, and Unclear. The diagrams on the left incorporate single-label assignments (self-loops) to represent the total prevalence of each category. The diagrams on the right depict only label co-occurrences and show the specific thematic intersections and overlapping logic identified by each model.

mial fit was unstable, the same linear predictor was estimated using a Poisson model.

We used cluster-robust standard errors³⁸ with clustering at the subreddit level and report incidence rate ratios with 95% confidence intervals for all focal terms. We also computed the predicted mean replies for combinations of M_i , HR_i , $Post_i$, and $Region_i$ by averaging fitted values over the observed covariate distribution.

Community-level harm reduction visibility and reply premium. To quantify harm reduction visibility, we aggregated comments to subreddit-week units. For each subreddit s and week w , we computed

$$HR_share_{s,w} = 100 \times \frac{\sum_{i \in (s,w)} \mathbf{1}\{HR_i = 1\}}{N_{s,w}},$$

where $N_{s,w}$ is the total number of comments in that subreddit-week. We then measured class-specific engagement with misconceptions using a reply premium on the log scale. Let r_i be the reply count for comment i and $y_i = \log(1 + r_i)$. For each class $c \in \{\text{Legal-status, Enforcement}\}$, we computed within each subreddit-week

$$\Delta_{s,w,c} = \bar{y}_{s,w,c}^{(1)} - \bar{y}_{s,w,c}^{(0)}, \quad \text{PremiumPct}_{s,w,c} = 100 \times \{\exp(\Delta_{s,w,c}) - 1\},$$

where the superscripts (1) and (0) denote class- c misconception comments and all other comments in the same cell. At least three comments in each group were required to compute $\text{PremiumPct}_{s,w,c}$.

Finally, we tested the association between harm reduction visibility and variation in reply premiums across subreddit-weeks. A weighted linear regression used $\text{PremiumPct}_{s,w,c}$ as the outcome and $HR_share_{s,w}$ as the main predictor, with indicators for region and misconception class included as covariates. The model weighted observations by $N_{s,w}$ and used robust standard errors for inference.

3 Results

We compared model outputs using label prevalence, label density, and pairwise normalized Hamming distance in the final analytic corpus of 12,537 comment records (Table 3). GPT-4o and Gemini-2.0-Flash placed most records in Label 5, with 80.40% and 84.13%, respectively, classified as unclear or containing no verifiable factual claim. Average label density was 1.05 for GPT-4o and 1.04 for Gemini-2.0-Flash, so most records received a single label.

LLaMA-3.2B produced a broader label profile, with 37.27% of records in Label 5 and an average label density of 1.52. Its mean distance from the other two models was 0.3247, compared with 0.1909 for GPT-4o and 0.1928 for Gemini-2.0-Flash. GPT-4o and Gemini-2.0-Flash were much closer to each other, with pairwise Hamming distance 0.0590. The same pattern appeared in the repeated-output check, where GPT-4o and Gemini-2.0-Flash returned stable label sets for most repeated row groups, while LLaMA-3.2B changed labels more often. GPT-4o was used as the primary classifier based on expert review and this more stable label profile.

Figure 2 visualizes label distribution and co-occurrence patterns across the three models. Co-occurrence refers to comments that received more than one substantive misconception label in a single prediction. The right-hand chords are sparse for GPT-4o and Gemini-2.0-Flash, which matches their average label density near 1.0. In contrast, LLaMA-3.2B shows broader overlap among Labels 1-4, which matches its higher label density and lower repeated-output stability. The dominant shared overlap for GPT-4o and Gemini-2.0-Flash links ‘Legal-Status Confusion’ with ‘Enforcement Misconceptions’. This pairing tracks debates about what is legal under the exemption and what police can still do.

Within GPT-4o, most co-occurrence mass concentrates in a small set of pairings in the right panel of Figure 2. The strongest

chord connects 'Legal-Status Confusion' and 'Enforcement Misconceptions', which aligns with comments that treat decriminalization as full legality and then infer that police powers stopped. A second visible chord links 'Policy-Target Misinterpretation' with 'Impact on Crime or Society', which captures comments that treat the policy goal as promoting use and then link that assumed goal to disorder. Other chords are thinner, which means GPT-4o rarely assigned extra labels when evidence for a second factual claim was weak.

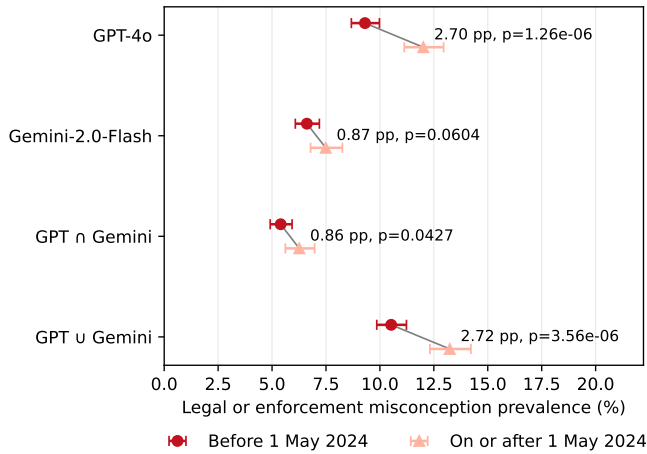


Figure 3: **Cross model sensitivity analysis for the legal or enforcement misconception indicator.** Points show the percentage of comment records classified as legal or enforcement misconception records before 1 May 2024 and on or after 1 May 2024. Error bars show 95% Wilson confidence intervals. The direction of change was positive under all four definitions and was statistically detectable under GPT-4o, the strict two model intersection, and the union definition.

3.1 Policy Misconception Themes

Reddit discussions of the BC policy contained recurring misconception themes after the no-claim category was excluded. In the final analytic corpus, GPT-4o assigned at least one substantive misconception label to 2,457 of 12,537 comment records (19.60%). The most common substantive labels were 'Impact on Crime or Society' (961, 7.67%), 'Legal-Status Confusion' (776, 6.19%), 'Enforcement Misconceptions' (694, 5.54%), and 'Policy-Target Misinterpretation' (649, 5.18%). As a sensitivity analysis using the labels most directly tied to documented policy rules, we combined *Legal Status Confusion* and *Enforcement Misconceptions* into a single legal or enforcement indicator. This indicator appeared in 1,299 of 12,537 comment records, or 10.36%. It increased from 712 of 7,650 records before 1 May 2024, or 9.31%, to 587 of 4,887 records after 1 May 2024, or 12.01% ($\chi^2_1 = 23.48$, $p = 1.26 \times 10^{-6}$). The change was mainly linked to Legal Status Confusion, which rose from 407 of 7,650 records before 1 May 2024, or 5.32%, to 369 of 4,887 records after 1 May 2024, or 7.55% ($p = 4.32 \times 10^{-7}$). *Enforcement Misconceptions* changed from 405 of 7,650 records, or 5.29%, to 289 of 4,887 records, or 5.91% ($p = 0.139$). GPT-4o and Gemini-2.0-Flash agreed on the combined legal or enforcement indicator for 11,804 of 12,537 records, or 94.15%,

with Cohen's $\kappa = 0.632$. In an unadjusted check, records with legal or enforcement labels had a mean of 0.97 direct replies, compared with 0.75 among other records.

As an additional cross model sensitivity check, we repeated the pre and post May 2024 comparison for the combined legal or enforcement indicator using four definitions: GPT-4o alone, Gemini-2.0-Flash alone, the intersection of GPT-4o and Gemini-2.0-Flash, and the union of GPT-4o and Gemini-2.0-Flash (Figure 3). GPT-4o increased from 712 of 7,650 pre revision records (9.31%) to 587 of 4,887 post revision records (12.01%), a difference of 2.70 percentage points ($p = 1.26 \times 10^{-6}$). Gemini-2.0-Flash showed the same direction, increasing from 506 of 7,650 records (6.61%) to 366 of 4,887 records (7.49%; $p = 0.060$). Under the strict intersection definition, records labelled by both models increased from 413 of 7,650 records (5.40%) to 306 of 4,887 records (6.26%; $p = 0.043$). Under the union definition, records labelled by at least one of the two models increased from 805 of 7,650 records (10.52%) to 647 of 4,887 records (13.24%; $p = 3.56 \times 10^{-6}$). The May 2024 increase was positive under all four definitions and was unlikely to be driven by a single classifier boundary.

Figure 4 illustrates the monthly composition of substantive label assignments after excluding Label 5. In most months, the composition was dominated by 'Impact on Crime or Society' and 'Policy-Target Misinterpretation', but the series also shows a mid-2024 shift toward legal scope and enforcement. In June 2024, legal-status or enforcement misconceptions appeared in 63 of 399 comment records (15.79%). In the same month, GPT-4o assigned 55 'Legal-Status Confusion' labels and 15 'Enforcement Misconceptions' labels. The increase was more visible in absolute volume and in reply engagement during the weeks after the May 2024 amendment. This uptick coincided with a significant policy amendment in May 2024, when Health Canada and the BC government narrowed the decriminalization pilot by prohibiting drug possession in all public spaces.³⁹ Since then, legal possession has been limited to private residences and certain designated sites only.³⁹ During that period, many Reddit users appeared unsure about what was actually allowed under 'decriminalization'. Some wondered if the policy had been effectively reversed and others complained that 'police are still arresting people', which suggests widespread confusion about the policy's legal boundaries and how enforcement would proceed after the amendment.

After July 2024, legality and enforcement labels became less prominent as discussion returned to the policy's targets and perceived societal impact. Their rise around the revision and decline in later months link the mid-2024 increase to the policy change rather than a lasting shift in the discourse.

3.2 Bursts of Reply Engagement

To determine when legal-status and enforcement misconceptions drew concentrated attention, we mapped the weekly reply premium for these records within each subreddit. Figure 5 presents the subreddit-week premium surface for the final analytic corpus. The burst procedure identified three premium bursts and no penalty bursts. Each cluster lasted across multiple weeks and appeared in multiple subreddits, which means

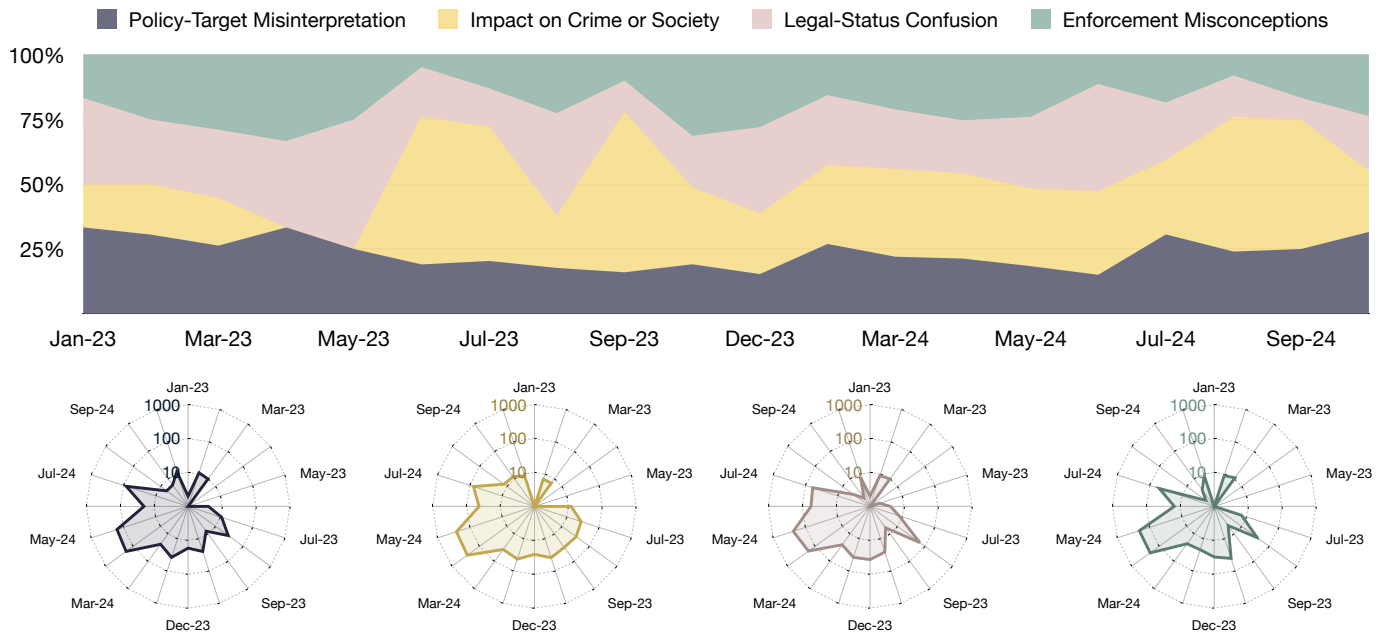


Figure 4: **Composition and dynamics of misconception themes over time (Jan 2023 to Sep 2024).** The top row is a percentage area chart based on monthly substantive label assignments in the final analytic corpus. Label 5 is excluded, and the four substantive labels are scaled to sum to 100% within each month. The bottom row presents radar plots of monthly label counts on a logarithmic radial scale, which emphasizes month-to-month intensity within each class and makes intermittent peaks more visible. For the radar plots, months with zero assignments are placed at the innermost ring for display. October 2024 is omitted because the collection window ends on 20 October 2024, which yields a partial month.

that the reply premium was associated with sustained public attention to legal scope and police authority.

The first burst occurred from March 25 to April 22, 2024 and involved *r/vancouver*, *r/canada*, and *r/VictoriaBC*. It spanned 5 weeks, included 7 hot cells, and had a mean reply premium of 22.9%. This period preceded the May 2024 policy revision, but it was already centered on the issues that later became the focus of the amendment. Comments during this window often treated visible public drug use as evidence that the exemption had removed meaningful limits on possession or police intervention. Our results show that local and national forums framed the issue at different scales. *r/vancouver* and *r/VictoriaBC* were closer to the daily visibility of public use and enforcement concern, while *r/canada* widened the discussion into a national debate about whether BC's policy represented a failure of decriminalization. In this sense, the first burst captured a pre-amendment period of pressure, when on-line discussion was already linking public disorder, legal permissibility, and police authority.

Two additional bursts began the week of 29 April 2024 and continued through 13 May 2024, overlapping the policy revision that removed the exemption for possession in public spaces. One burst involved *r/VictoriaBC*, *r/CanadaPolitics*, *r/britishcolumbia*, and *r/ontario*. It spanned 3 weeks, included 5 hot cells, and had a mean reply premium of 38.1%, the largest mean premium among the three bursts. This cluster covers the first weeks when the legal boundary changed and users debated what decriminalization still meant after public space possession was no longer exempt. The presence of *r/CanadaPolitics* and *r/ontario* shows that the amendment was not only a local BC issue. It

also became a national policy story about the limits of decriminalization and the return of enforcement in public settings.

A second burst during the same 29 April to 13 May window involved *r/toronto*, *r/saskatchewan*, and *r/saskatoon*. It spanned 3 weeks, included 4 hot cells, and had a mean reply premium of 21.0%. This cluster occurred outside the main BC local forums. In these communities, BC's experience was used as a reference point in broader debates about drug policy, public safety, and municipal enforcement. The same policy change generated two linked discussion patterns. One concentrated in BC and policy focused communities, and the other spread through non BC forums where users debated if similar policies should be adopted or avoided elsewhere.

Our results showed no penalty bursts. Legal status and enforcement misconception records periodically drew more replies than other records in the same subreddit week, but there was no comparable multi-week pattern in which they drew fewer replies. This asymmetry indicates that legal scope and police authority became recurring points of argument during the amendment period. Figure 5 shows where those arguments concentrated. The highest reply-premium cells identify the weeks and communities where short clarification messages would have been most timely. At those points, brief explanations could have distinguished decriminalization from legalization, clarified where the possession exemption applied and what enforcement authority remained, and directed readers to harm reduction services. Posting this information in active threads would have placed accurate policy guidance where discussion was most concentrated.

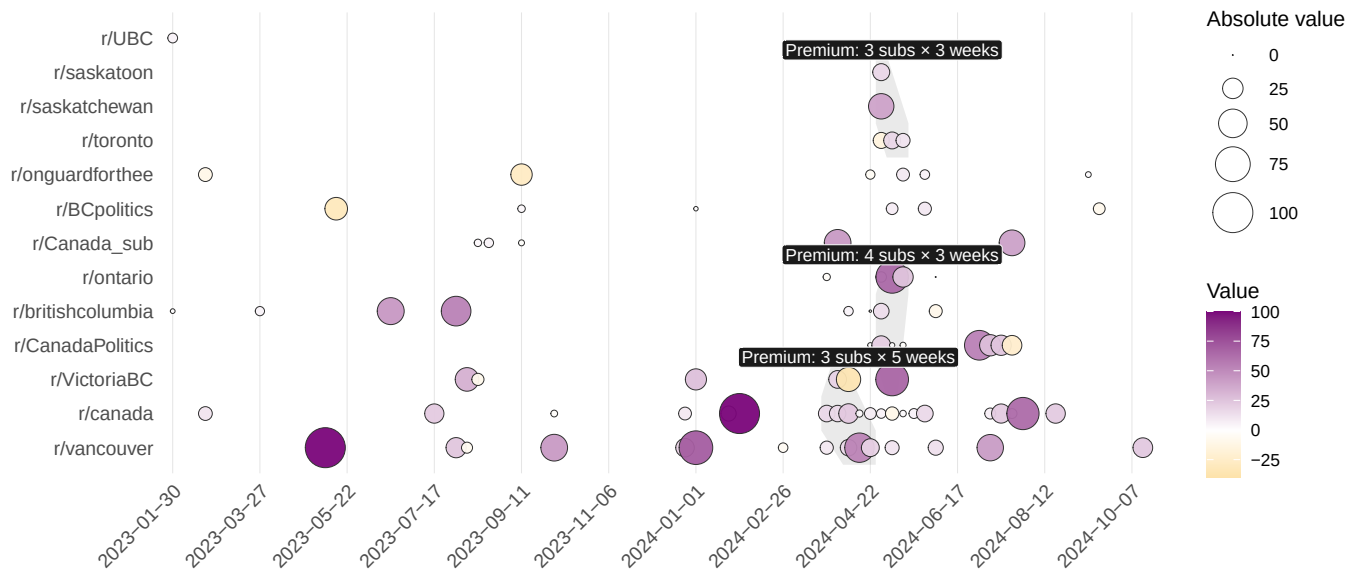


Figure 5: **Weekly reply premium by subreddit with multi-week $\times$ multi-subreddit bursts.** Points are subreddit-week observations from the final analytic corpus. Point area encodes [premium], and color is centered at zero. Purple indicates a positive premium for legal-status or enforcement misconception records, and orange indicates a negative premium. Polygons mark the three premium bursts. For display, color and point area are capped at 100 percentage points, and burst detection used the uncapped premium values.

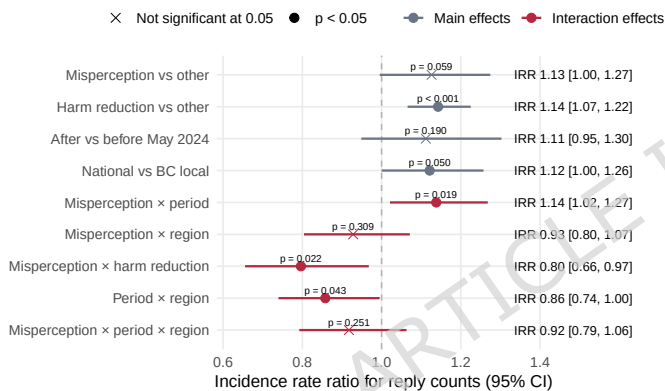


Figure 6: **Incidence rate ratios (IRR) for direct reply counts from a negative binomial regression.** Points show incidence rate ratios and bars show 95% confidence intervals. The model includes comment length, upvotes, hour, weekday, and subreddit fixed effects. Interaction terms link legal or enforcement misconceptions with policy period, region, and harm reduction content

3.3 Harm reduction discourse and engagement with legal/enforcement misconceptions

Figure 6 summarizes adjusted associations between legal or enforcement misconceptions, harm reduction content, policy period, region, and direct reply counts. Legal or enforcement misconception records had a higher adjusted reply-rate estimate than other records, although the main effect had limited precision (IRR 1.13, 95% CI 1.00-1.27, $p = 0.059$). Harm reduction content had a stronger reply premium, with an IRR of 1.14 (95% CI 1.07-1.22, $p < 0.001$). Reply rates after May 2024 were estimated to be 11% higher than before, but this association was not statistically significant (IRR 1.11, 95% CI 0.95-1.30, $p = 0.190$). National records also had an estimated 12% higher reply rate than BC local records (IRR 1.12, 95% CI

1.00-1.26, $p = 0.050$).

The legal or enforcement misconception $\times$ period term was positive, with IRR 1.14 (95% CI 1.02-1.27, $p = 0.019$). This means that the added reply activity linked to legal or enforcement misconceptions was stronger after the May 2024 revision. The misconception $\times$ region term was close to null (IRR 0.93, 95% CI 0.80-1.07, $p = 0.309$), which means the average misconception premium was similar in BC local and national subreddits after adjustment. The period $\times$ region term was below one (IRR 0.86, 95% CI 0.74-1.00, $p = 0.043$), suggesting smaller post-revision increases in BC local communities than in national ones. The three-way misconception $\times$ period $\times$ region term had limited precision (IRR 0.92, 95% CI 0.79-1.06, $p = 0.251$).

The misconception $\times$ harm reduction interaction was below one (IRR 0.80, 95% CI 0.66-0.97, $p = 0.022$), but this does not mean that comments mentioning harm reduction received fewer replies because harm reduction content had a positive main effect. Instead, the additional reply premium associated with a legal or enforcement misconception was smaller when the same comment also mentioned harm reduction. These exchanges may have focused more on information, service references, or clarification, even though both topics attracted replies.

Figure 7 links subreddit-week harm reduction visibility to the class-specific reply premium for enforcement misconceptions and legal-status confusion. For the two classes combined, the volume-weighted ordinary least-squares slope was positive, but the estimate was uncertain. A 10 percentage-point higher harm reduction share was associated with a 4 percentage-point higher reply premium (95% CI -1.72 to 9.72 , $p = 0.174$, $R^2 = 0.07$). The association depended on the misconception class, with a near zero slope for enforcement misconceptions of -0.25 percentage points for each 10 percentage-point increase

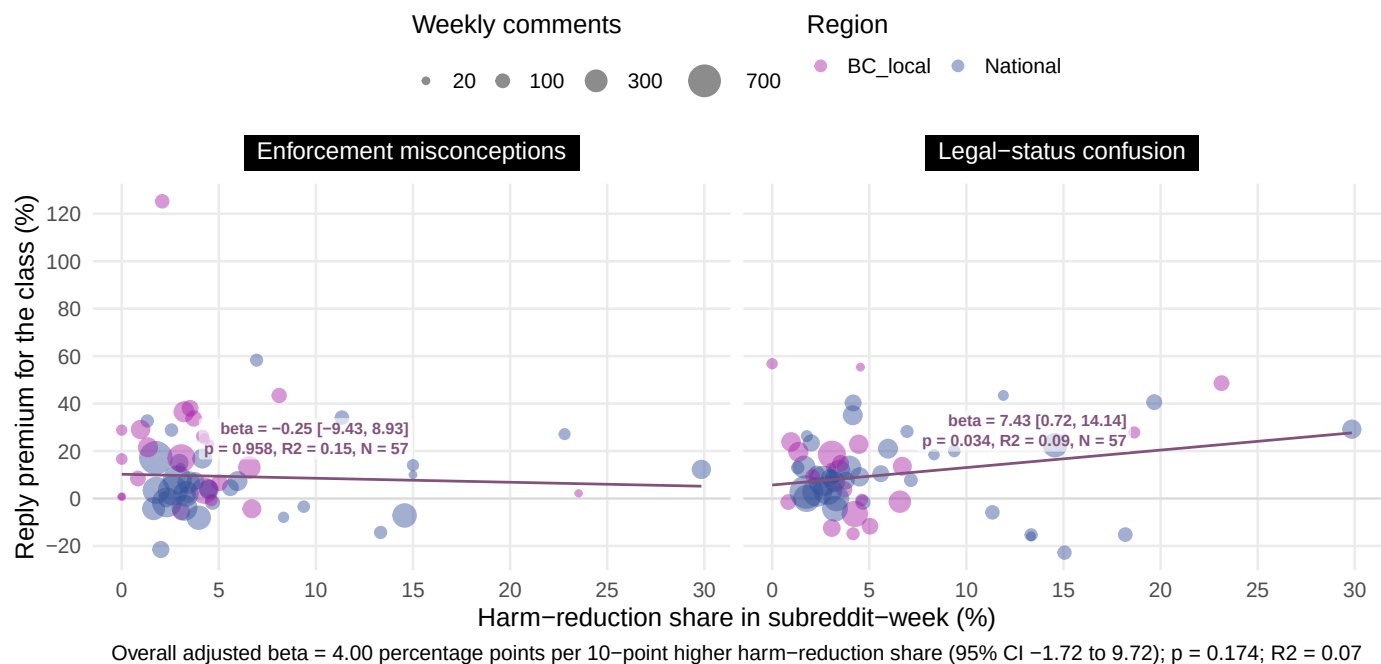


Figure 7: **Harm reduction visibility and reply premium by misconception class.** Each point represents a subreddit-week from the final analytic corpus. The horizontal axis gives the percentage of comments in that subreddit-week that referenced harm reduction, and the vertical axis gives the class-specific reply premium for enforcement misconceptions or legal-status confusion. Point size shows weekly comment volume, and color distinguishes BC local from national communities. Lines are volume-weighted ordinary least squares fits within each class.

in harm reduction share (95% CI -9.43 to 8.93, $p = 0.958$, $R^2 = 0.15$, $N = 57$) and a positive slope for legal-status confusion of 7.43 percentage points (95% CI 0.72 to 14.14, $p = 0.034$, $R^2 = 0.09$, $N = 57$). Harm reduction visibility did not consistently reduce legal or enforcement debate at the community level, as weeks with more harm reduction discussion had a larger reply premium for legal-status confusion but not for enforcement misconceptions.

4 Discussion

4.1 Online Misconceptions as Implementation Signals

Reddit comments provide a record of the parts of the pilot that drew public debate as the rules changed. In our study, users repeatedly returned to three practical questions: what decriminalization permitted, what authority police retained, and how visible public drug use should be interpreted. These questions correspond to the legal-status, enforcement, and impact labels that were prominent in the quantitative results. To interpret these patterns, we reviewed model disagreement cases and comments that received many replies alongside the aggregate findings. The examples below show how these concerns appeared in discussion and add context to the quantitative results.

Perceived Disorder and Public Safety: Concern about public safety often relied on the visibility of drug use in shared spaces. Many commenters treated visible public use as direct evidence that decriminalization increased crime and overdoses, as one commenter wrote: “*Since decriminalization was put in place, there has been more open drug use. More social disorder and crime. I get all of those things like mental health and*

housing availability and people's income but decriminalization made things worse.” The same concerns also appear in national survey data. For instance, the *Canadians' Knowledge and Attitudes Around Drug Decriminalization* survey reported that 51% of respondents in 2023 and 53% in 2024 believed decriminalization would lead to greater harms, such as overdoses.¹⁹ Similarly, 43% in 2023 and 48% in 2024 agreed that decriminalizing drugs would make their communities less safe.¹⁹

Enforcement data show a different pattern, with criminal enforcement for personal possession declining soon after decriminalization. During the initial exemption period in BC, possession offences and sub-threshold seizures fell sharply from pre-policy levels.^{40,41,42,43,44} Fewer offences and seizures meant fewer criminal justice consequences for personal possession. Findings from other jurisdictions also do not support attributing concurrent increases in overdose deaths directly to decriminalization. In Oregon and Washington, quasi-experimental analyses found no clear increase in overdose deaths over the first year of decriminalization, while arrests for possession declined substantially.^{13,45} In Oregon, models that accounted for the rapid spread of fentanyl no longer attributed overdose mortality to decriminalization.⁴⁶ Qualitative research from the same period documented fentanyl's growing dominance in local markets. This change in the drug supply may help explain rising fatalities independently of the legal reform.⁴⁷ Decriminalization may change the visibility of drug use and public expectations, while overdose deaths are more closely tied to the toxic drug supply and social conditions than to legal status.^{43,13,46}

Misunderstandings About Enforcement and Consequences: Many comments treated decriminalization as the end

of penalties for possession and police intervention. One commenter wrote: “*The concept that someone can be fined for having a drink in public but you can do drugs... with no consequences makes no sense.*” National survey data show that this view was held by a majority of respondents. The share who believed that any drug could be possessed legally and that police no longer stopped people carrying illegal substances increased from 60% in 2023 to 65% in 2024.¹⁹ Research with people who use drugs also found uncertainty about how decriminalization operated in practice. Participants sometimes treated decriminalization and legalization as the same policy and were unsure when police could still intervene.²⁴ This uncertainty could lead some people to assume that enforcement had ended and others to expect continued police contact. Either interpretation may make it harder to understand when or how to seek services. The most recent *Quarterly Decriminalization Data* report documents continued police enforcement. More than half of possession offences recorded between May 2024 and January 2025 occurred in public spaces such as streets, roads, highways, and parking lots.⁴⁸ Clear communication during implementation should explain that decriminalization and legalization are separate legal approaches and state what authority police retain.

Confusion About Permissible Locations: Comments also questioned where possession or use was permitted. Many commenters assumed that decriminalization covered all public spaces and were especially concerned about places used by children. One commenter wrote: “*This was so stupid. Why would you ever make that legal in public places? There’s a time and place for everything. Smoking meth at the bus stop in front of kids is not it.*” These comments overstated where the exemption applied. The original exemption in January 2023 already prohibited possession in certain spaces, and amendments in September 2023 expanded restrictions in child-focused areas.⁴⁸ A revised exemption in May 2024 narrowed permitted settings further and limited possession to private residences, legal sheltering sites, and supervised consumption services.⁴⁸ Frequent changes to the exemption likely increased confusion for the public and for people most exposed to enforcement, which increases opportunities for misinformation to circulate. Public guidance should name the locations directly. A brief notice could state where possession remained exempt and where police could still intervene. Specific information about location rules may help people avoid police contact caused by misunderstanding the exemption.

Moral Framing and Policy Purpose: A separate theme involved moral interpretations of the policy’s intent. Some commenters framed decriminalization as government endorsement of drug use, instead of a measure aimed at reducing stigma and linking people with care, as one commenter wrote: “*The government encouraging drug use has been extremely triggering as someone trying to stay away from it.*” Media framing provides context for this response. A recent scoping review reports that media portrayals often characterize people who use drugs as *criminal*, *violent*, or *dangerous*, which can shift policy preferences toward punitive approaches rather than treatment.⁴⁹ Opioid-related coverage also often focuses on scenes of public overdoses and the presence of police or first responders, which

can frame the crisis through a criminal justice lens.⁵⁰

When its public health purpose and available services are poorly explained, decriminalization may look like government approval of drug use. Public communication could cite the reported increase in the proportion of people who use drugs accessing overdose prevention and supervised consumption services.⁴⁸ The policy also depends on people being able to find and use treatment and harm reduction services. Portugal removed criminal penalties while investing in treatment and social support. In Portugal, entry into services includes an assessment of needs such as housing and family support. Harm reduction programs also provide supplies and information about treatment options.⁵¹

Engagement Bursts and Communication Timing: Replies to legal status and enforcement misconceptions increased around the May 2024 amendment. In our final analytic corpus, premium bursts appeared from late March through mid-May 2024, with two bursts beginning the week of April 29 and continuing through May 13. These bursts occurred as possession in public spaces was removed from the exemption and discussion turned to what police could still do. At the same time, comments that misstated the scope of the exemption drew longer reply chains. Disputed claims about the revised rules repeatedly prompted argument.

Administrative records after the May 2024 revision show that, in the following eight months, a large share of recorded possession offences in British Columbia involved public locations where possession had become criminalized again.^{25,39} The overlap in timing provides context for the rise in replies about legal status and police authority. A short post could have named the locations still covered by the exemption and stated where police could intervene. A second line could have directed readers to harm reduction services. The subreddit by week map in Figure 5 shows where and when these comments received more replies than surrounding discussion. Those cells mark the active conversations in which a brief explanation of the revised rules may have been most useful.

4.2 Harm Reduction Legitimacy in Online Debate

Evidence of effectiveness is only one part of harm reduction policy. Public debate about safety, responsibility, and policy intent can influence continued support for naloxone distribution and supervised consumption services, both of which have demonstrated benefits.^{52,53,54}

Public opinion data indicate a broad base of support for harm reduction in Canada. A national survey found that about 64% of Canadians support harm reduction approaches.⁵⁵ Earlier research in Canada, the UK, and Australia also reported majorities backing needle programs and supervised consumption.⁵⁶ Even with majority support, policy commitment has often remained unstable. In one systematic review, public health experts noted that survey findings of strong support were ‘ignored by some policy-makers and media’, alongside negative portrayals of safer drug use practices.⁵⁶ Canadian policy analyses have described some harm reduction frameworks as largely rhetorical, with limited implementation.⁵⁵ Studies of these debates also find that moral and emotional commitments can carry

more weight than effectiveness data.⁵⁷ Selective media coverage can influence public attitudes and policy preferences over time. Cultivation processes link sustained exposure to recurring narratives with changes in social reality beliefs, particularly when individuals have limited direct experience with an issue.⁵⁸ Recent analyses of Canadian news similarly found that coverage of youth substance use often emphasized generalized 'harms on the rise' and individual-level solutions, with limited attention to social context.⁵⁹ Repeated coverage of visible disorder may push public debate toward crime and safety, even when evidence supports harm reduction interventions.^{53,52}

Online forums can circulate practical harm reduction advice and stigmatizing or inaccurate claims at the same time.⁶⁰ Health communication scholarship describes this dynamic as part of an 'infodemic', where misleading information can contribute to confusion, risk-taking behaviours, and reduced trust in health authorities.^{61,62} Word choice can also affect policy preferences. In a randomized study with clinicians, exposure to the label 'substance abuser' increased endorsement of punitive responses compared with describing the person as having a 'substance use disorder'.⁶³ This result shows how blame-focused language can move discussion toward punishment and away from care.

Posts about harm reduction need plain explanations of the policy limits and language that avoids stigma. Lived experience accounts can make harm reduction less abstract. Brief explanations can define decriminalization as a separate legal approach, name the locations covered by the exemption, and state what authority police retain. These messages are most useful before misleading claims spread widely. Inoculation and prebunking approaches can help readers recognize common misleading frames in short digital formats.^{64,65} For opioid policy, agencies could post brief explanations of the current rules before periods of high attention and add corrections when false claims begin drawing replies.^{61,62}

4.3 Operationalizing Rapid Harm Reduction and Early Warning on Social Platforms

Our study used reply activity to locate periods when misconceptions drew concentrated attention in opioid policy discussions. In our data, comments containing legal or enforcement misconceptions received more replies, with the largest increase after the May 2024 revision. Harm reduction references were also associated with higher reply counts.^{25,39,48}

Two findings help interpret what this engagement means for harm reduction implementation directly. Comments that mentioned harm reduction received more replies, suggesting that some threads were used to exchange practical information. When harm reduction terms appeared in comments containing legal or enforcement misconceptions, the added reply premium was smaller. Practical information may have redirected some exchanges toward clarification. Prior forum research describes similar norms, where users share dosing cautions, safer use practices, and mutual aid during disruptive periods.^{36,37} We then considered the same relationship at the subreddit week level.

The burst analysis showed how these patterns appeared across communities and weeks. Two bursts overlapped the May

revision that changed the rules for possession in public spaces. Before the revision, comments about legal scope and police authority were already drawing replies in local and national communities. During the revision window, those comments became more concentrated across subreddits. Discussion during this period centred on overdose risk, visible drug use, and uncertainty about the revised rules. Users questioned where possession remained exempt and what police could do in public settings.

The comment level and subreddit week results were different. The pooled association between harm reduction visibility and reply premium was positive but imprecise, while the class-specific panels separated two dynamics. Enforcement misconceptions had a near-zero relationship with harm reduction visibility. Legal-status confusion had a positive slope, which means weeks with more harm reduction discussion were also weeks when confusion about legal status drew more replies. Harm reduction discussion and legal uncertainty often appeared in the same high attention weeks. The legal rules and overdose risk operated through different mechanisms. Decriminalization changes how personal possession is handled by police. Overdose risk remains closely tied to the toxic drug supply, housing conditions, treatment access, and harm reduction services. Evaluations in Oregon and Washington found large declines in possession arrests after decriminalization, while overdose trends were more closely linked to fentanyl market dynamics than to enforcement levels.^{13,45,46,47} This difference helps explain the Reddit bursts. A policy revision can make legality and police authority more confusing at the same time that overdose risk remains tied to supply and service conditions. Communication should not treat these as separate audiences. It should explain the legal boundary and point people toward naloxone, drug checking, supervised consumption, opioid agonist treatment, and safer supply information in the same message. Portugal shows how legal reform and service access can be introduced together. Its reform paired reduced criminal penalties with treatment and support services.⁵¹ Our Reddit results suggest a similar need online: explain the legal rules and make service information easy to find. When legal-status confusion rises, service signposting should not be delayed until after misinformation spreads. Short posts in active threads can state where possession remains exempt, where it does not, what police authority remains, and where people can access harm reduction supports. This is especially important during amendment periods, when the legal boundary changes faster than public understanding.

Agencies could use public platform discussions as one part of implementation monitoring under ethical social listening standards.⁶⁶ Communication should track exemption revisions and explain location rules and remaining enforcement powers.^{48,25,39} Prebunking that distinguishes decriminalization from legalization can prepare readers before high attention bursts online.^{64,65} Partnerships with moderators can keep naloxone, opioid agonist treatment, supervised consumption, and safer supply resources visible when new threads form.^{36,37} Agencies can also connect online signals to offline response planning by reviewing indicators and adjusting outreach or service information when confusion rises.^{67,68} This treats online

discussion as one source of implementation feedback, alongside public notices, service delivery, and local outreach.

4.4 Limitations

Several limitations should be considered when interpreting the online signals captured in this study and when generalizing beyond Reddit. The corpus was constructed from a hosted export of Reddit threads returned by the phrase 'BC Decriminalization'. This search strategy may miss relevant discussions that used different wording, appeared in other languages, or occurred in communities not captured by the query. Activity was also concentrated in large subreddits, especially *r/canada*, *r/vancouver*, and *r/britishcolumbia*. The findings should not be interpreted as population views in British Columbia or Canada. They describe comment-level discourse within the identified Reddit threads.

The study also focused on comments, not all text linked to a Reddit thread. Post titles and metadata were used to identify entry points into relevant discussions, but post bodies, self-post text, link-post titles, linked news articles, and article comment sections were not analyzed as observations. This design matches the engagement outcome, since the models use direct replies to comments. At the same time, it may miss policy explanations, harm reduction terms, or framing cues introduced in the original post or linked article. Our findings should be read as patterns in comment level discussion, not as a full analysis of all text connected to each Reddit thread.

Our data processing rule collapses exact repeated export rows, but it does not merge near-duplicate comments, copied text posted at different times, or similar comments from different subreddits. Usernames, user identifiers, profile links, and account histories were not kept in the analytic files and were not used in any analysis. This protected privacy, but it also prevented evaluation of bots, repeated accounts, or geographic location.

The repeated export rows were useful for checking model stability. They gave multiple model outputs for the same comment text, subreddit, timestamp, reply count, and upvote count. Across 9,578 repeated row groups, GPT-4o returned the same parsed label set in 9,057 groups (94.56%) and a different parsed label set in 521 groups (5.44%). Gemini-2.0-Flash returned the same parsed label set in 9,244 groups (96.51%) and a different parsed label set in 334 groups (3.49%). LLaMA-3.2B returned the same parsed label set in 3,057 groups (31.92%) and a different parsed label set in 6,521 groups (68.08%). These repeated-output patterns were consistent with the model comparison in the main Results. GPT-4o and Gemini-2.0-Flash behaved more consistently under the taxonomy, and LLaMA-3.2B varied more often for the same comment record. For final analysis, each repeated row group was assigned one conservative comment-level label set from the existing model outputs.

The classification task also has conceptual limits. We limited the misconception taxonomy to four checkable domains and one no-claim category. This choice gave a clear structure for large-scale annotation, but it also means that the estimates do not capture every important feature of the discussion.

Stigma, moral framing, fear, sarcasm, local frustration, and uncertainty were assigned Label 5 when they did not include a checkable claim about policy purpose, legal status, enforcement powers, or direct causal impact. Zero-shot classification with general-purpose models can also misread sarcasm, idiom, and meanings that depend on earlier comments in a thread. Manual review of disagreement cases helped select the primary model, but this review was a stress test for model choice and not a corpus-wide estimate of error. Some labels still require judgment about evidentiary strength, especially when comments refer to news reports, visible public use, or personal experience.

The boundary between written policy, enforcement practice, and public perception is another limitation. Some comments appeared to refer to the legal text of the exemption. Others referred to local enforcement, visible public drug use, or a mix of both. As these are different reference points, the legal status and enforcement labels should be read as signals of confusion about policy rules and police authority, not as proof that a commenter misunderstood daily implementation. For outcome claims, Label 4 captures comments that present the exemption as the direct, sole, or settled cause of crime, overdose deaths, or public disorder. It does not capture all negative views of the policy. This distinction reduces overclassification, but judgment cannot be fully removed from the task.

The harm reduction measure was also conservative. We identified harm reduction content through explicit terms related to naloxone, drug checking, supervised consumption, overdose prevention, syringe services, opioid agonist treatment, safer supply, and relevant provincial institutions. This approach supports reproducibility, but it may miss informal language, indirect references, or comments that discuss care and safety without using these terms. It also does not measure the quality, accuracy, or usefulness of the harm reduction information shared in a comment.

The reply-premium and burst analyses identify when legal-status or enforcement misconception records attracted more direct replies than other records in the same subreddit-week. They cannot separate correction from amplification. A high reply premium may indicate disagreement, clarification, argument, stigma, or support. Direct reply counts also do not capture voting dynamics, deeper thread structure, private messages, cross-platform diffusion, or later offline effects. Burst timing may be influenced by external news, moderation practices, local events, or policy announcements that were not modeled as time-varying covariates. The May 2024 revision was used as an implementation reference point, but causal attribution is limited in observational social media data.

Finally, data acquisition used a hosted third-party interface available at the time of collection. Changes in Reddit search, platform content, or third party settings may prevent the same hosted export from being collected again. The archived files and code allow the exported data to be processed into the same comment records, labels, figures, and statistical models. Future work can extend the collection window, include additional search terms and platforms, analyze post-level framing as a separate corpus, and link online signals to administrative and health indicators on finer time scales.

5 Conclusion

We analyzed 12,537 Reddit comment records to quantify how policy misunderstanding concentrates attention during decriminalization implementation in British Columbia. Zero-shot multi-label classification mapped each comment to a fixed misconception taxonomy and identified impact, legal-status, enforcement, and policy-purpose misconceptions as recurring themes. Repeated export rows provided an internal model stability check before one conservative label set was assigned to each final comment record. GPT-4o and Gemini-2.0-Flash were stable across repeated outputs, while LLaMA-3.2B varied more often. Legal and enforcement confusion increased after the May 2024 amendment, with the largest concentration in June 2024. The adjusted reply model found a higher but imprecise main effect for legal or enforcement misconception records, and a stronger post-May reply premium. Weekly reply premiums clustered into three multi-subreddit premium bursts from late March to mid-May 2024, with two bursts overlapping the policy revision period. Harm reduction references attracted replies on their own, and their co-occurrence with legal or enforcement misconceptions was linked to a smaller added reply premium. At the community level, harm reduction visibility had an imprecise pooled association with reply premiums, with a positive class-specific relationship for legal-status confusion and a near-null relationship for enforcement misconceptions. These findings support time-sensitive public communication during policy change. Messages should state place-based possession rules, clarify remaining police authority, and link readers to naloxone, supervised consumption, drug checking, and opioid agonist treatment. Burst detection can help agencies identify when and where confusion is concentrating online, which can guide brief clarification and service signposting in the communities where debate is active.

Abbreviations

BC	British Columbia
AB	Alberta
ON	Ontario
LLM	Large Language Model
IRR	Incidence Rate Ratios
CI	Confidence Interval
SD	Standard Deviation
IQR	Interquartile range

Author's contributions

ZS conceptualized and designed the study, conducted the data analysis and visualization, interpreted the results, and wrote the final version of the manuscript. MS, CPY, KS, and MN contributed to data collection and study conceptualization. SH contributed to the implementation and validation of the LLM pipeline. MS contributed to the implementation and validation of the data labelling process, writing of the first draft,

and interpretation of the results. KS, CD, and MN reviewed, edited, and provided feedback on the manuscript. ZS was responsible for funding acquisition and overall supervision of the project.

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Data availability

The data used in this study, including the archived raw export and the labeled datasets produced by each large language model used in the Methods, are publicly available in the provided GitHub repository.²³ The repository also includes a data acquisition metadata file that documents the hosted actor name, collection pathway, query string, known input settings, output fields, and fields removed before analysis. The acquisition step is documented as provenance for the archived export. It is not presented as local scraper code. Since Reddit search behavior, platform content, and third party tool settings can change, exact future recreation of the same live export is not claimed.

Declarations

Ethics approval and consent to participate

This study analyzed publicly available Reddit comments and did not involve direct interaction with human participants. Because this research used open data and included no identifiers, ethics approval and participant consent were not required.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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